

02L2-Intro

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December 18, 2024

Slide 1

Content

CS 24300: AI Basics

Tony Bergstrom

Slides adapted from Yexiang Xue and Tony Bergstrom's original set

Description

This slide is a title slide for a lecture on Artificial Intelligence Basics, likely part of a course called CS 24300. It displays the course title, the instructor's name (Tony Bergstrom), and a credit line indicating that the slides are adapted from another source (Yexiang Xue and Tony Bergstrom). The overall presentation style is a simple, dark gray background with light gray text. The context is crucial for understanding the content of the lecture, which will likely cover fundamental concepts in AI.

Figures

Slide 2

Content

Announcements

Initial Lab - this week is on your own. Play with AI. Not in person, upload to Gradescope.
Discussion post introduction

Description

This slide displays announcements for a class, likely an Artificial Intelligence Basics course. It specifically mentions an "Initial Lab" that students are to complete independently this week. The lab is to be done using AI tools, and the results are to be uploaded to a platform called Gradescope. A discussion post is also announced, likely to be introduced after the lab. The overall tone is instructional and informative, providing necessary details for the upcoming activities.

Figures

Slide 3

Content

Today's lecture

Introduce the basic idea of AI

- History
- The nature of intelligence
- Characteristics of the field of study

Description

This slide, likely from an Artificial Intelligence Basics course, outlines the topics to be covered in the current lecture. The title "Today's lecture" clearly indicates the purpose of the slide. The bullet points under "Introduce the basic idea of AI" list the specific subtopics: History, The nature of intelligence, and Characteristics of the field of study. This indicates that the lecture will provide an overview of AI, including its historical context, the concept of intelligence itself, and the key aspects of the field. The bullet points are a standard way to present a list of topics in a lecture format.

Figures

Slide 4

Content

The beginnings of AI Research

1956 Dartmouth summer research project

"We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire.

The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves."

Description

This slide, from a lecture on Artificial Intelligence Basics, details the 1956 Dartmouth Workshop, a pivotal event in the history of AI. It presents a quote from the workshop's proposal, outlining the goals and motivations behind the project. The text describes a 2-month, 10-person study focused on artificial intelligence, aiming to simulate learning and other cognitive functions in machines. The proposal emphasizes the belief that these aspects of intelligence can be precisely described and replicated by machines. The quote also mentions the ambition to enable machines to use language, form abstractions, solve human-level problems, and even improve themselves. The slide likely serves as an introduction to the historical context of AI research, highlighting the early aspirations and foundational ideas that shaped the field. Images of John McCarthy, Marvin Minsky, and Claude Shannon are included, likely to represent key figures involved in the Dartmouth Workshop.

Figures



(a)



(b)



(c)



(d)

Slide 5

Content

What is artificial intelligence?

[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning Bellman 1978

Computational Intelligence is the study of the design of intelligent agents. Poole 1988

The study of how to make computers do things at which, at the moment, people are better Rich Knight 1991

The study of computation that make it possible to perceive, reason, and act. Winston 1992

Description

This slide, from a lecture on Artificial Intelligence Basics, defines artificial intelligence. It presents several different perspectives on AI, each attributed to a specific author and year. The first definition, enclosed in square brackets, describes AI as the automation of activities associated with human thinking, including decision-making, problem-solving, and learning. This is followed by a definition of Computational Intelligence as the study of designing intelligent agents. Subsequently, the slide presents the study of how to make computers perform tasks better than humans currently can, and finally, the study of computation enabling perception, reasoning, and action. Each definition provides a different angle on the concept of AI, highlighting different aspects of the field. The years listed after each definition indicate when these perspectives were proposed, providing historical context for the development of AI thought.

Figures

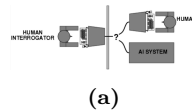
Slide 6

Content

Description

This slide, from an Artificial Intelligence Basics course, discusses the Turing Test. It presents the core idea of the test, which is to determine if a machine can exhibit intelligent behavior. The slide states that the question "Can machines think?" is equivalent to "Can machines behave intelligently?". It further clarifies that the Turing Test is an operational test for intelligent behavior, specifically framed as the "Imitation Game." Crucially, the slide emphasizes that there are no constraints on the topics that can be used in the test. A diagram illustrates the setup of the test. A human interrogator interacts with both a human and an AI system, trying to determine which is which based on their responses. The interrogator's inability to distinguish between the human and the AI indicates that the AI has exhibited intelligent behavior. The diagram shows the human interrogator on the left, the AI system in the middle, and a second human on the right. An arrow connects the interrogator to the AI system, labeled with a question mark, signifying the interrogator's attempt to distinguish between the two. The context of the slide is within a broader discussion of the historical and philosophical foundations of AI, highlighting the Turing Test as a significant milestone in the field.

Figures



Slide 7

Content

Turing Test: Inspiring a Field

Does it mean machines can think if they pass the Turing test?

John Searle: Chinese room argument in "Minds, Brains, and Programs"

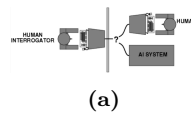
[draw, circle, minimum size=1cm] (human1) at (0, 1) HUMAN; [below of=human1, node distance=1cm] (interrogator) INTERROGATOR; [draw, rectangle, minimum width=2cm, minimum height=1.5cm] (ai) at (3, 1) AI SYSTEM; [draw, circle, minimum size=1cm] (human2) at (6, 1) HUMAN;

[->] (interrogator) -- (ai); [->] (interrogator) -- (human2); [->] (ai) -- (interrogator); [above, midway between=interrogator and ai] (question) ?;

Description

This slide from an Artificial Intelligence Basics lecture discusses the Turing Test and its implications. It presents the question of whether passing the Turing Test implies machine thought. The slide highlights John Searle's "Chinese room" argument, a critique of the Turing Test's ability to determine genuine understanding in machines. A diagram illustrates the Turing Test setup. A human interrogator interacts with both a human and an AI system, trying to determine which is which based on their responses. The interrogator's inability to distinguish between the human and the AI indicates that the AI has exhibited intelligent behavior. The diagram shows the human interrogator on the left, the AI system in the middle, and a second human on the right. An arrow connects the interrogator to the AI system, labeled with a question mark, signifying the interrogator's attempt to distinguish between the two. The context of the slide is within a discussion of the philosophical and practical aspects of AI, specifically exploring the limitations of the Turing Test as a measure of machine intelligence.

Figures



Slide 8

Content

Turing's Prediction

"I believe that in about fifty years' time it will be possible to programme computers, with a storage capacity of about 10^9 , to make them play the imitation game so well that an average interrogator will not have more than 70 per cent. chance of making the right identification after five minutes of questioning..."

"...The original question, 'Can machines think?' I believe to be too meaningless to deserve discussion. Nevertheless I believe that at the end of the century the use of words and general educated opinion will have altered so much that one will be able to speak of machines thinking without expecting to be contradicted..."

"Computing Machinery and Intelligence", Mind, 1950, p. 442

Description

This slide presents a quote from Alan Turing's seminal paper "Computing Machinery and Intelligence," published in Mind in 1950. It details Turing's prediction about the future of artificial intelligence. Specifically, Turing predicts that within fifty years, computers with a storage capacity of approximately 10^9 will be able to convincingly simulate human conversation in the imitation game (the Turing Test). He further states that the average person will have less than a 70

Figures

Slide 9

Content

Aspects of AI: Research tracks at AAAI-2020

Causality
Search, Constraint Satisfaction, Optimization
Game Theory and Economic Paradigms
Computer Vision Natural Language Processing
Machine Learning
Knowledge Representation and Reasoning
AI for Social Good
Multi Agent Systems
Reasoning Under Uncertainty
Robotics
Planning/Scheduling

Description

This slide from an Artificial Intelligence Basics course lists research tracks in AI, specifically those presented at the AAAI-2020 conference. It categorizes various aspects of AI research into different themes. The categories include Causality, Search/Constraint Satisfaction/Optimization, Game Theory/Economic Paradigms, Computer Vision/Natural Language Processing, Machine Learning, Knowledge Representation/Reasoning, AI for Social Good, Multi-Agent Systems, Reasoning Under Uncertainty, Robotics, and Planning/Scheduling. The context of the slide is within a broader discussion of current trends and areas of focus in AI research, providing a snapshot of the diverse topics being explored in the field.

Figures

Slide 10

Content

AlphaGo Beat world champion Lee sedol in 2016

Description

This slide displays a picture of the historic 2016 match between AlphaGo, a Google DeepMind AI, and world champion Go player Lee Sedol. The image shows the match in progress, with the players seated at a Go board. The title clearly indicates the outcome of the match, highlighting a significant milestone in the development of artificial intelligence, specifically in the domain of game playing. The image also includes logos and text indicating the event as a Google DeepMind Challenge Match, held between March 8th and 15th, 2016. The context of the course (Artificial Intelligence Basics) is crucial for understanding the slide's relevance, as it showcases a key moment in the demonstration of AI's capabilities in complex strategic games.

Figures



(a)

Slide 11

Content

Search over Game Trees

Rules define set of legal moves, each move changes board

Sequence of moves forms a game tree of configurations; Leaf nodes represent outcomes (with associated score)

Algorithms search over game tree to learn which moves, in which configurations, are likely to lead to wins
 [scale=0.7] [circle, draw, fill=red!20] (root) at (0,0) ×; (child1) at (-1.5,-1.5) ×; (child2) at (0,-1.5) ×; (child3) at (1.5,-1.5) ×; (root) – (child1); (root) – (child2); (root) – (child3); [below right] at (0,-2.5) Win +1; [below left] at (-1.5,-2.5) Lose -1; [below right] at (1.5,-2.5) Draw 0;

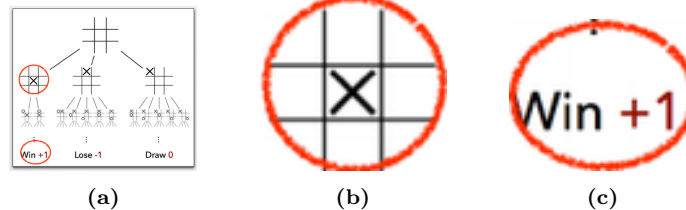
Until the end of the game

Game tree: Traced by Stannered, original by Gdr (Image:Tic-tac-toe-game-tree.png) [GFDL (<http://www.gnu.org/copyleft/fdl.html>), via Wikimedia Commons]

Description

This slide, from an Artificial Intelligence Basics course, discusses "Search over Game Trees." It explains that rules define legal moves in a game, and a sequence of moves creates a game tree. Leaf nodes in this tree represent game outcomes, each with an associated score (e.g., Win +1, Lose -1, Draw 0). Algorithms traverse this tree to identify moves and configurations that are likely to lead to winning outcomes. A simple example of a game tree, likely Tic-Tac-Toe, is shown. The root node (a circled X) represents the initial game state. The branches emanating from the root represent possible first moves. The leaf nodes at the bottom of the tree represent the final outcomes of those moves. The slide also includes labels indicating the scores associated with each outcome. The context of the course is crucial, as it demonstrates a fundamental AI technique for decision-making in games.

Figures



Slide 12

Content

Search Tree Sizes

Tic-tac-toe: 10^5

Checkers: 10^{31}

Chess: 10^{123}

Backgammon: 10^{144}

Go: 10^{360}

Atoms in your body $\sim 10^{28}$

Atoms in the universe $\sim 10^{80}$

Assume that a computer evaluates 1 million board configurations per second.

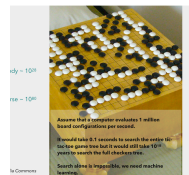
It would take 0.1 seconds to search the entire tic-tac-toe game tree but it would still take 10^{18} years to search the full checkers tree.

Search alone is impossible, we need machine learning.

Description

This slide, from an Artificial Intelligence Basics course, presents the vastness of search spaces in various games. It lists the estimated number of possible game configurations (search tree sizes) for Tic-Tac-Toe, Checkers, Chess, Backgammon, and Go. The numbers are expressed in exponential notation (10^n). The slide also includes comparisons to the number of atoms in the human body and the universe, highlighting the astronomical scale of these search spaces. Crucially, the slide emphasizes that even with a fast computer, searching the entire game tree for complex games like checkers is computationally infeasible. The slide concludes by stating that brute-force search alone is insufficient for these games, and machine learning techniques are necessary to overcome the computational limitations.

Figures



(a)

Slide 13

Content

Description

This slide, from an Artificial Intelligence Basics course, presents three examples of AI applications. The left image shows a self-driving car, likely a Google car, with the Google logo visible. The middle image depicts a futuristic smart building, with digital data or binary code patterns integrated into the architectural design. The right image shows a person receiving personalized medicine, with a medical professional administering an injection. The slide title clearly indicates the three areas of application. The images are accompanied by a caption that includes the source and license information for the images, indicating that they are from Wikimedia Commons and are licensed under Creative Commons. The context of the course is crucial, as it demonstrates how AI is being applied in various sectors, including transportation, architecture, and healthcare.

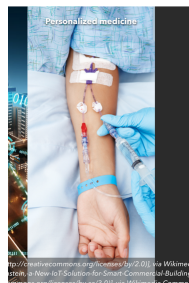
Figures



(a)



(b)



(c)

Self-driving cars

(d)

Smart buildings

(e)

Slide 14

Content

We've made great progress in AI over the last 60 years...
But we've also notoriously underestimated problem difficulty...

Description

This slide, likely from an Artificial Intelligence (AI) lecture, makes a statement about the progress made in AI over the past 60 years. It acknowledges significant advancements but also highlights the tendency to underestimate the inherent complexities of AI problems. The text is a concise summary, likely intended to introduce or conclude a discussion about the challenges and progress in the field of AI.

Figures

Slide 15

Content

In 1966, Seymour Papert (from the same MIT AI group as McCarthy and Minsky) told his graduate students to "solve" computer vision as a summer project unlike many other problems in AI, vision was tractable and "easy" to encode Decades later, we are still working...

Description

This slide from an Artificial Intelligence Basics course discusses a historical anecdote about computer vision. It states that in 1966, Seymour Papert, a member of the MIT AI group, assigned computer vision as a summer project to his graduate students. The slide highlights that, in contrast to other AI problems, computer vision was perceived as relatively straightforward and easily encodable at that time. The concluding statement, "Decades later, we are still working...", implies that despite this initial optimism, significant challenges remain in the field of computer vision even today.

Figures

Slide 16

Content

Facebook's visual system can automatically identify components of images and produce verbal description for the blind

Description

This slide, likely from an Artificial Intelligence Basics course, showcases an application of computer vision. It depicts a Facebook visual system that automatically identifies components within images and generates a verbal description. The slide contains a grid of images, each with colored outlines highlighting different objects or components within the scene. These outlines suggest that the system is segmenting and classifying the objects in the images. The text below the images states that the system can produce verbal descriptions for the blind, highlighting the practical application of this AI technology. A URL is included, linking to an article likely discussing Yann LeCun and Facebook's AI research. The context of the course is crucial, as it demonstrates a specific application of AI in image processing and natural language generation.

Figures



(a)

Slide 17

Content

Surprises brought by the development of Artificial Intelligence...

Description

This slide, likely from an Artificial Intelligence Basics course, presents a title that suggests a discussion about unexpected or surprising outcomes resulting from advancements in AI. The phrase "Surprises brought by the development of Artificial Intelligence..." implies a focus on the unanticipated consequences or phenomena that have emerged as AI has progressed. The lack of specific details suggests that this is an introductory or transitional slide, setting the stage for a discussion about the topic.

Figures

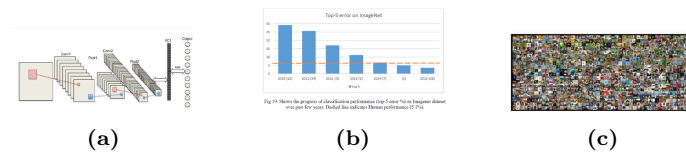
Slide 18

Content

Description

This slide from an Artificial Intelligence Basics course presents a visual representation of a convolutional neural network (CNN). The title, "Neural Net Achieves Super Human Performance," indicates the focus on the performance of a neural network, likely a CNN, in image recognition tasks. The slide shows a diagram of a CNN, with labeled layers: Conv1, Pool1, Conv2, Pool2, FC1, Output, and Loss. Arrows depict the flow of data between layers. The diagram illustrates the sequential processing of image data through convolutional and pooling layers, followed by a fully connected layer and a loss function. The slide also includes a graph showing the progress of classification performance (top-5 error rate) on the ImageNet dataset over several years. The graph shows a decreasing trend in error rate, indicating improved performance over time. A dashed horizontal line represents the human performance level (5.1).

Figures



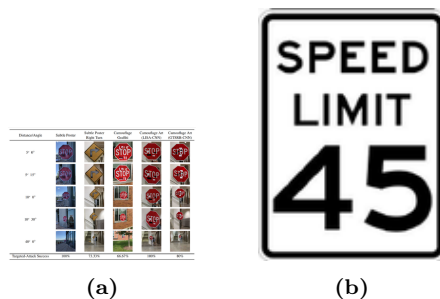
Slide 19

Content

Description

This slide, likely from a computer vision or deep learning research paper (CVPR 2018), presents results of a study on the robustness of deep learning models to physical-world attacks. The title "AI safety" suggests the focus on the potential vulnerabilities of AI systems in real-world applications. The table displays a grid of images, each representing a different type of attack on a "STOP" sign. The columns represent different types of attacks: "Subtle Poster," "Subtle Poster Right Turn," "Camouflage Graffiti," "Camouflage Art (LISA-CNN)," and "Camouflage Art (GTSRB-CNN)." Each row corresponds to a different distance/angle at which the attack is applied. The images show variations in the appearance of the "STOP" sign, making it harder for the AI system to recognize it. The final row shows the success rate of the targeted attack for each type of attack. The values (e.g., 100

Figures



Slide 20

Content

Ethical issues of artificial intelligence

nature

NEWS 24 OCTOBER 2019 UPDATE 26 OCTOBER 2019

Millions of black people affected

by racial bias in health-care algorithms

Study reveals rampant racism in decision-making software used

by US hospitals - and highlights ways to correct it.

Description

This slide presents a news article headline from the journal "nature" about ethical issues in artificial intelligence. The title, "Ethical issues of artificial intelligence," indicates the focus of the article. The article details how millions of black people have been affected by racial bias in health-care algorithms used by US hospitals. The article highlights the rampant racism in decision-making software and suggests ways to correct it. The date of the article (October 24, 2019, updated October 26, 2019) is also included. The context of the course (Artificial Intelligence Basics) is crucial for understanding the ethical implications of AI systems and the potential for bias in real-world applications. This slide likely serves as an introduction to a discussion about ethical considerations in AI development and deployment.

Figures



(a)

Slide 21

Content

Playing Games

Description

This slide, likely slide 21 of a 50-slide presentation, simply displays the title "Playing Games" in a large, bold font. The context of the course, "Artificial Intelligence Basics," suggests that this is the beginning of a section or module dedicated to game playing as a specific application of AI. The gray background and lack of further content indicate that this is a title slide or a transition slide, setting the stage for the following content on game playing within the context of AI.

Figures

Slide 22

Content

Playing Games

Description

This slide, slide 22 of a 50-slide presentation, displays a single title: "Playing Games." The title is in bold, sans-serif font. The background is dark gray or black. There is no other content on the slide. The context of the course, "Artificial Intelligence Basics," suggests that this slide is a heading for a section or module about game playing as an application of AI. It's likely the start of a discussion or presentation on AI techniques used in game development, game playing strategies, or the use of games as a benchmark for AI development.

Figures



(a)

Slide 23

Content

Minimax

Description

This slide, slide 23 of a 50-slide presentation, displays the single word "Minimax" in a large, bold font. The background is a video frame, showing a person giving a lecture. The context of the course, "Artificial Intelligence Basics," suggests that this is the title of a section or module about the minimax algorithm in game theory, a crucial concept in AI for game playing and decision-making under uncertainty. The slide is likely the introduction to a discussion of the minimax algorithm, its application in various games, and its relationship to AI strategies.

Figures



(a)

Slide 24

Content

How do you play games: Algorithm components

- Representation of the state of the game
- Set of actions available
- Planning sequences of actions (look ahead)
- Reasoning about opponent's possible actions
- How to choose between actions?

Description

This slide, slide 24 of a 50-slide presentation on Artificial Intelligence Basics, lists the components of an algorithm for playing games. The title, "How do you play games: Algorithm components," clearly indicates the topic. The bullet points below the title detail the key elements involved in designing an AI game-playing agent. These include: representing the current state of the game, identifying the available actions, planning sequences of actions (often involving a look-ahead strategy), considering the opponent's possible actions, and finally, determining the best action to take. The context of the course suggests that this slide is a prelude to a discussion of specific algorithms and techniques used in game playing AI, such as minimax, alpha-beta pruning, or Monte Carlo Tree Search.

Figures

Slide 25

Content

Autonomous robots

Description

This slide, slide 25 of a 50-slide presentation on Artificial Intelligence Basics, displays a single title: "Autonomous robots." The title is in bold, sans-serif font. The background is a light gray. This slide likely serves as a heading for a section or module dedicated to the topic of autonomous robots within the context of AI. It's a prelude to a discussion of AI techniques and algorithms used to create robots capable of operating independently, potentially including topics like path planning, navigation, object recognition, decision-making, and interaction with the environment.

Figures

Slide 26

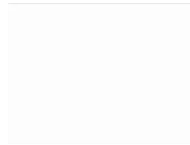
Content

Blank Slide

Description

This slide, slide 26 of a 50-slide presentation on Artificial Intelligence Basics, is completely blank. There is no text, images, or other content. It is likely a transition slide or a placeholder for a later addition. The context of the course suggests that this slide is not part of the current lecture's content.

Figures



(a)

Slide 27

Content

How is soccer different from chess?

Description

This slide, slide 27 of a 50-slide presentation on Artificial Intelligence Basics, poses a question: "How is soccer different from chess?". The question is presented in a bold, sans-serif font, likely as a lead-in to a discussion comparing the complexities and characteristics of these two games from an AI perspective. The context of the course suggests that this slide is a transition to a section or module that will explore the differences in game representation, possible actions, and the challenges of developing AI agents for each game. The slide is likely to follow a discussion of game playing algorithms and introduce the concept of different game characteristics that influence the design of AI strategies.

Figures

Slide 28

Content

Algorithm components

- Representation of the state of the game
- Set of actions available
- Planning sequences of actions (look ahead)
- Reasoning about opponent's possible actions
- How to choose between actions?

Description

This slide, slide 28 of a 50-slide presentation on Artificial Intelligence Basics, lists the components of an algorithm for a game, likely a game like soccer or chess, from an AI perspective. The title "Algorithm components" indicates that the slide is discussing the different parts of an AI algorithm designed to play the game. The bullet points detail the key elements: representing the current game state, identifying available actions, planning sequences of actions (looking ahead), considering the opponent's possible actions, and finally, the crucial decision-making process of choosing the best action. The context of the course suggests that this slide is a prelude to a discussion of specific algorithms or techniques used for game playing AI, such as minimax, alpha-beta pruning, or Monte Carlo Tree Search. The question "How to choose between actions?" highlights the core challenge of developing an effective AI agent for such games.

Figures

Slide 29

Content

Speech synthesis

Description

This slide, slide 29 of a 50-slide presentation on Artificial Intelligence Basics, displays the title "Speech synthesis" in a bold, sans-serif font. The slide is entirely text-based, with no images or other content. It is likely a heading or a title slide for a section of the presentation that will discuss the topic of speech synthesis in AI. The context of the course suggests that this slide is a transition to a module or section that will explore the techniques and methods used to create synthetic speech, potentially including topics such as text-to-speech conversion, speech synthesis algorithms, and the challenges involved in creating natural-sounding speech. The slide is a simple, clear indicator of the topic to follow.

Figures

Slide 30

Content

No content visible on slide 30

Description

Slide 30 of a 50-slide presentation on Artificial Intelligence Basics shows no content. The slide is completely blank, likely a placeholder or an error in the presentation. The context of the course is irrelevant as there is no information to describe.

Figures



(a)

Slide 31

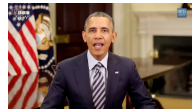
Content

No content visible on slide 31

Description

Slide 31 of a 50-slide presentation on Artificial Intelligence Basics shows a video of Barack Obama giving a speech. There is no text or other content on the slide itself. The slide is entirely a video display. The context of the course is irrelevant as there is no information to describe beyond the visual content.

Figures



(a)

Slide 32

Content

How is speech synthesis similar to game playing?

Description

This slide, slide 32 of a 50-slide presentation on Artificial Intelligence Basics, poses a question: "How is speech synthesis similar to game playing?". The slide contains only this question in a bold, sans-serif font, likely serving as a prompt for discussion or a transition to a section comparing the two AI tasks. The context of the course suggests that the presentation is exploring the similarities and differences between various AI applications, potentially highlighting commonalities in problem-solving approaches, search strategies, or the use of data and models. The question implies that the presenter intends to explore the underlying computational processes and strategies involved in both speech synthesis and game playing, and how these might be analogous.

Figures

Slide 33

Content

Algorithm components

- Representation of the state of the game
- Set of actions available
- Planning sequences of actions (look ahead)
- Reasoning about opponent's possible actions
- How to choose between actions?

Description

Slide 33 of a 50-slide presentation on Artificial Intelligence Basics lists the components of an algorithm for game playing. The slide title is "Algorithm components". The bullet points detail the key elements necessary for an AI to play a game, including: representing the current game state, identifying available actions, planning sequences of actions, considering the opponent's possible moves, and finally, selecting the best action. The context of the course suggests that this slide is part of a section discussing game playing algorithms, likely focusing on the strategic and decision-making aspects of AI agents in competitive environments. The "look ahead" component highlights the importance of planning and anticipating future consequences of actions.

Figures

Slide 34

Content

Conversational agents

Description

Slide 34 of a 50-slide presentation on Artificial Intelligence Basics displays the title "Conversational agents". The slide contains only this title, written in a bold, sans-serif font, likely indicating the beginning of a section dedicated to conversational AI agents. The context of the course suggests that the presentation is exploring different types of AI applications, and this slide signals a transition to a discussion of agents designed to engage in human-like conversations. No further details are visible on the slide.

Figures

Slide 35

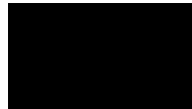
Content

No content visible

Description

This slide, slide 35 of a 50-slide presentation on Artificial Intelligence Basics, is completely blank. There is no text, images, or other content visible. This likely indicates a transition to a new section or a placeholder for future content.

Figures



(a)

Slide 36

Content

Algorithm components

Representation of knowledge about the world
Set of decisions (actions)
Reasoning about opponents or effects of actions
Need to choose between actions

- How to choose?
- Generalize to new situations?

Description

Slide 36 of a 50-slide presentation on Artificial Intelligence Basics lists the components of an algorithm, specifically for game playing or problem-solving agents. The title is "Algorithm components". The bullet points detail the key elements necessary for an AI agent to function, including: representing the current state of the world, identifying available actions, considering the opponent's possible actions or the effects of its own actions, and the crucial step of selecting the best action. The slide also includes questions about how to choose the best action and how to generalize the learned strategies to new situations. The context of the course suggests that this slide is part of a section discussing game playing algorithms or general problem-solving agents, focusing on the strategic and decision-making aspects of AI agents in competitive or complex environments.

Figures

Slide 37

Content

Fraud detection

Description

Slide 37 of a 50-slide presentation on Artificial Intelligence Basics displays the title "Fraud detection". This likely signals the beginning of a section dedicated to AI applications in fraud detection. The slide contains only this title, written in a bold, sans-serif font. The context of the course suggests that the presentation is exploring various AI applications, and this slide indicates a transition to a discussion of fraud detection systems and the use of AI in identifying fraudulent activities.

Figures

Slide 38

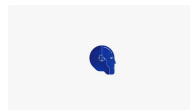
Content

No content visible

Description

Slide 38 of a 50-slide presentation on Artificial Intelligence Basics shows a graphic of a stylized, profile view of a robot head in a dark blue color. There is no text or other content on the slide. This likely indicates a transition to a new section or a visual break, possibly to introduce a new topic or concept in the presentation. The context of the course suggests that the presentation is covering fundamental AI concepts, and this slide is likely not directly related to the previous content on fraud detection.

Figures



(a)

Slide 39

Content

Algorithm components

Representation of knowledge about the world
Set of decisions (actions)
Reasoning about opponents or effects of actions
Need to choose between actions

- How to choose?
- Generalize to new situations?

Description

Slide 39 of a 50-slide presentation on Artificial Intelligence Basics lists the key components of an algorithm, likely for game-playing or problem-solving agents. The title is "Algorithm components". The bullet points detail the necessary elements for an AI agent to function: representing the current state of the world, identifying available actions, considering the opponent's possible actions or the effects of its own actions, and the crucial step of selecting the best action. The slide also includes questions about how to choose the best action and how to generalize the learned strategies to new situations. The context of the course suggests that this slide is part of a section discussing game playing algorithms or general problem-solving agents, focusing on the strategic and decision-making aspects of AI agents in competitive or complex environments.

Figures

Slide 40

Content

Driving cars

Description

Slide 40 of a 50-slide presentation on Artificial Intelligence Basics displays the title "Driving cars". This likely signals the beginning of a section dedicated to AI applications in autonomous vehicle technology. The slide contains only this title, written in a bold, sans-serif font. The context of the course suggests that the presentation is exploring various AI applications, and this slide indicates a transition to a discussion of AI-powered self-driving car systems and the related algorithms and technologies.

Figures

Slide 41

Content

No content visible

Description

Slide 41 of a 50-slide presentation on Artificial Intelligence Basics shows a completely blank, black slide. There is no text, images, or other content. This likely indicates a transition to a new section or a visual break, possibly to introduce a new topic or concept in the presentation. The context of the course suggests that the presentation is covering fundamental AI concepts, and this slide is likely not directly related to the previous content on autonomous vehicles.

Figures



(a)

Slide 42

Content

Algorithm components

Representation of knowledge about the world
Set of decisions (actions)
Reasoning about opponents or effects of actions
Need to choose between actions

- How to choose?
- Generalize to new situations?

Description

Slide 42 of a 50-slide presentation on Artificial Intelligence Basics lists the key components of an algorithm, likely for game-playing or problem-solving agents. The title is "Algorithm components". The bullet points detail the necessary elements for an AI agent to function: representing the current state of the world, identifying available actions, considering the opponent's possible actions or the effects of its own actions, and the crucial step of selecting the best action. The slide also includes questions about how to choose the best action and how to generalize the learned strategies to new situations. The context of the course suggests that this slide is part of a section discussing game playing algorithms or general problem-solving agents, focusing on the strategic and decision-making aspects of AI agents in competitive or complex environments. The bullet points are formatted with a leading dot, indicating a list of items.

Figures

Slide 43

Content

Why WERE autonomous vehicles chosen as AN AI Grand Challenge?

Description

Slide 43 of a 50-slide presentation on Artificial Intelligence Basics poses the question "Why WERE autonomous vehicles chosen as AN AI Grand Challenge?". The slide contains only this question, written in a bold, sans-serif font. This suggests that the presentation is discussing the reasons behind selecting autonomous vehicle technology as a significant benchmark or challenge in the field of artificial intelligence. The context of the course suggests that the presentation is exploring various AI applications, and this slide is likely introducing a section dedicated to the specific challenges and opportunities presented by autonomous vehicles as a testbed for AI development.

Figures

Slide 44

Content

THE AI STACK

- Machine Learning, Deep Networks
- Search, Plan and Prove
- Carnegie Mellon University
- Machine Learning
- Autonomy
- Sensors, Computer Vision, Computer Audio
- Human-AI Interaction

Description

Slide 44 of a 50-slide presentation on Artificial Intelligence Basics displays a diagram representing "The AI Stack." The diagram is centered on a stylized image of a brain, depicted as a network of interconnected points and lines (a wireframe or mesh representation). The brain graphic is teal or turquoise in color against a black background. The title "THE AI STACK" is prominently displayed above the graphic. Various components or layers of AI are connected to the brain graphic, represented by text labels. These labels include "Machine Learning, Deep Networks," "Search, Plan and Prove," "Carnegie Mellon University Machine Learning," "Autonomy," "Sensors, Computer Vision, Computer Audio," and "Human-AI Interaction." The layout suggests a hierarchical or interconnected structure of AI components, with the brain as a central representation of the overall AI system. The context of the course suggests that this slide is part of a section discussing the different components and areas of focus within AI, potentially highlighting the interconnectedness of these areas and the role of Carnegie Mellon University in AI research.

Figures



(a)

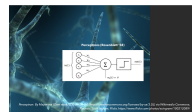
Slide 45

Content

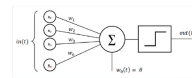
Description

Slide 45 of a 50-slide presentation on Artificial Intelligence Basics displays a diagram of a perceptron, a fundamental artificial neural network. The title "Perceptrons (Rosenblatt '58)" identifies the model. The diagram shows a network with multiple input nodes ($x_1, x_2, x_3, \dots, x_n$), each connected to a summation node (Σ). Each connection has a weight ($W_1, W_2, W_3, \dots, W_n$). The summation node calculates a weighted sum of the inputs. This sum is then passed through a step function (depicted by a large square symbol $\square$), which outputs either 1 or 0, representing the perceptron's output ($out(t)$). A bias term, $w_0(t) = \theta$, is also included in the calculation. The diagram visually represents the basic structure of a perceptron, illustrating how it takes multiple inputs, applies weights, sums them, and produces a binary output. The context of the course suggests that this slide is part of a section introducing fundamental concepts in neural networks, likely focusing on the perceptron as a foundational model for more complex neural networks.

Figures



(a)



(b)

Slide 46

Content

Reinforcement Learning

Checkers Program (Samuel '59)

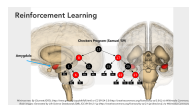
[scale=0.5] [circle, draw, fill=red, inner sep=1pt] (8) at (0,0) 8; [circle, draw, fill=red, inner sep=1pt] (13) at (2,2) 13; [circle, draw, fill=red, inner sep=1pt] (17) at (4,2) 17; [circle, draw, fill=red, inner sep=1pt] (1) at (-2,-2) 1; [circle, draw, fill=red, inner sep=1pt] (6) at (-1,-4) 6; [circle, draw, fill=red, inner sep=1pt] (11) at (0,-4) 11; [circle, draw, fill=red, inner sep=1pt] (15) at (2,-4) 15; [circle, draw, fill=red, inner sep=1pt] (25) at (4,-4) 25; [circle, draw, fill=red, inner sep=1pt] (22) at (3,-6) 22; [circle, draw, fill=red, inner sep=1pt] (27) at (5,-6) 27;

(8) - (13); (8) - (1); (8) - (6); (13) - (17); (13) - (11); (13) - (15); (17) - (22); (17) - (25); (17) - (15); (6) - node[below] NIL (1); (6) - node[below] NIL (-1,-5); (11) - node[below] NIL (0,-5); (15) - node[below] NIL (2,-5); (25) - node[below] NIL (4,-5); (22) - node[below] NIL (3,-7); (27) - node[below] NIL (5,-7);

Description

Slide 46 of a presentation on Artificial Intelligence Basics, specifically focusing on Reinforcement Learning, displays a graphical representation of a minimax tree. The tree visually depicts the game tree for the Checkers Program (Samuel '59). Nodes in the tree are circles, filled with red, and labeled with numerical values (e.g., 8, 13, 17). These numbers likely represent evaluation scores or utility values at each state in the game. The branches connecting the nodes are labeled with "NIL," indicating that these are possible moves or states in the game. The structure of the tree shows the possible game states and their associated evaluations. The slide also includes a 3D brain image in the background, which suggests a connection between the reinforcement learning algorithm and the human brain. The context of the course is Reinforcement Learning, and the slide is illustrating a specific application of the algorithm to the Checkers game, highlighting the evaluation process and possible moves within the game tree. The numbers on the nodes likely represent the evaluation of the state, and the NIL labels indicate the lack of a specific move or state.

Figures



(a)



(b)

Slide 47

Content

AI Success: Game Playing

How do basic Game Playing systems work?

Description

Slide 47 of a 50-slide presentation on Artificial Intelligence Basics, titled "AI Success: Game Playing," poses a question: "How do basic Game Playing systems work?" The slide contains only the title and the question, indicating that this is likely an introductory slide to a section discussing game playing within AI. The context of the course suggests that the following slides will delve into the methods and techniques used by AI systems to play games, potentially covering topics such as search algorithms, evaluation functions, and game tree analysis.

Figures

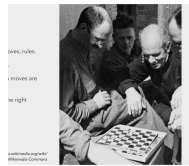
Slide 48

Content

Description

Slide 48 of a presentation on Artificial Intelligence Basics, focuses on the topic of game playing. The slide title is "Game Playing." The content describes a method for learning game strategies. It suggests learning about the game's components (board, pieces, moves, rules). The method involves watching and/or playing many games to observe outcomes and identify "good" moves. From these observations, strategies are developed. The slide concludes with a question, prompting the audience to consider whether the described method represents a correct strategy. The inclusion of a historical image of people playing checkers suggests a focus on the historical context of game playing and the development of strategies in AI. The URL and filename at the bottom of the slide indicate a source for the image, likely a picture of people playing checkers.

Figures



(a)

Slide 49

Content

AI Success: game playing

1994: Chinook checkers program (UAlberta) declared world champion
1995: TD-Gammon backgammon program (IBM) achieved grandmaster level,
**learned to play opening positions differently than humans*
1997: Deep Blue chess program (IBM) beat world chess champion Gary Kasparov,
**much of success was due to brute force search*
2016: AlphaGo Go program (Google) beat 9-dan professional player Lee Sedol,
**move 37: no human would have played, but turned tide of game and resulted in a surprising loss for Sedol*

Description

This slide, titled "AI Success: game playing," details milestones in AI game playing. It lists achievements of various programs in different games. For each program, the year of its achievement and the program's name (e.g., "Chinook checkers program") are given. The slide also includes key details about the programs' successes, such as the Chinook program being declared world champion, the TD-Gammon program achieving grandmaster level, and Deep Blue beating Garry Kasparov. Crucially, the slide highlights that Deep Blue's success was largely due to brute-force search. Finally, the AlphaGo program's victory over Lee Sedol is noted, with emphasis on a specific move (move 37) that was considered unexpected by human players. The asterisks ("*") likely indicate important details or further information about the achievements. The context of the course (Artificial Intelligence Basics) suggests that the slide is discussing the progress and methods used in AI game playing, highlighting significant achievements and the evolution of techniques.

Figures



(a)

Slide 50

Content

AlphaGo System

Deep learning + Search

One model predicts next move, given current state of board, trained on 30 million positions from human games

Another model predicts likelihood of winning given current state, trained on 30 million positions from self-play

System combines two models using Monte Carlo search

Silver, David, et al. "Mastering the game of Go with deep neural networks and tree search." Nature 529.7587 (2016): 484-489.

Description

This slide, titled "AlphaGo System," details the architecture and methodology behind the AlphaGo program. It describes a two-model approach using deep learning and search algorithms. The first model predicts the next optimal move given the current game state, trained on a dataset of 30 million positions from human Go games. The second model predicts the likelihood of winning from the current game state, also trained on 30 million positions, but this time from self-play games between AlphaGo instances. The system combines these two models using a Monte Carlo search algorithm to make its final move decisions. The slide also includes a citation to the original research paper by Silver et al. (2016) in Nature, which details the AlphaGo system's success in defeating a professional Go player. The slide likely follows a section on game playing in the course "Artificial Intelligence Basics," highlighting a specific example of a successful AI system in a complex game.

Figures

